Supplementary Materials

Continuous Grip-Force Dynamics as an Online Biophysical Signal of Perceptual Strain

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Table of contents

Main manuscript: manuscript.qmd (HTML/PDF) · manuscript-biorxiv.qmd (bioRxiv preprint PDF)

1 Signal processing and feature extraction

Grip-force preprocessing followed the pipeline documented in the main Methods and in `analysis/01_ingest.qmd`:

1. Target-normalized error: $e(t) = (F(t) - F_{\text{target}})/F_{\text{target}}$
2. Linear detrending within each trial epoch
3. Fourth-order zero-phase Butterworth high-pass filter at 0.5 Hz (`sosfiltfilt`)
4. Welch PSD estimation (Hann window, 50% overlap, ~0.5 Hz resolution)
5. Band integration: visuomotor tracking (0.5–3.0 Hz) and physiological tremor (8.0–12.0 Hz)

Epochs spanned grip-cue onset through 1 s after probe onset. All spectral features were log-transformed before modeling.

2 Model specifications

Mixed-effects logistic regression. Perceptual accuracy was modeled as:

$$\text{logit}\, P(\text{correct}) = \beta_0 + \beta_1 \text{modality} + \sum_{k=1}^{4} \beta_k \text{poly}_k(\text{difficulty}) + \beta_5 z(\text{tracking}) + \beta_6 z(\text{tremor}) + u_i$$

where $u_i \sim \mathcal{N}(0, \sigma^2)$ is a random intercept for participant i, and $z(\cdot)$ denotes within-condition standardized band power. Primary inference used high-grip trials only (40% MVC); low-grip models used the same structure.

Cross-validated machine learning. Random-forest and logistic classifiers were trained with subject-wise cross-validation (leave-one-subject-out) on high-grip trials. Feature sets compared:

- **Baseline:** stimulus difficulty level, task modality
- **DSP:** baseline features plus tracking- and tremor-band power

Performance was evaluated using area under the ROC curve (AUC).

Table 1: Cross-validated AUC for perceptual accuracy prediction (high-grip trials).

Feature set	Model	AUC	N trials
Baseline	Logistic	0.6170	7001
Baseline	RandomForest	0.8498	7001
DSP	Logistic	0.6279	6911
DSP	RandomForest	0.8553	6911

3 Sample coverage and attrition

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

Stage	Subject-tasks	Trials	% retainable	Note
Oddball subject-tasks in behavioral v3	136	18,099	—	aud + vis; runs merged into trial counts
Excluded: aborted mid-task	13	478	—	beh_trials < 100

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

Stage	Subject-tasks	Trials	% retainable	Note
Retainable subject-tasks	123	17,621	100.0%	individual missed trials already removed in v3
No grip files (entire subject×task)	1	128	0.7%	zero grip CSVs found; all behavioral trials in block excluded
Partial grip (behavioral > on-disk grip)	36	3,678	20.9%	some grip files exist but fewer trials than behavioral (often 1 run 30 trials)
Grip trials on disk	122	14,430	81.9%	fatigue-break runs merged per session; retainable subject-tasks only
Feature extraction loss	95	671	3.8%	grip trials present but spectral features not extracted (short epochs, etc.)
Analysis set (trial features extracted)	122	13,759	78.1%	primary grip force modeling cohort

4 Additional figures and tables

Single-trial signal chain. Representative high-grip incorrect trial showing the transformation from raw force to normalized error, rolling variance, and Welch PSD with analysis bands marked.

Probe-locked spectrogram. Time–frequency power aligned to probe onset ($t = 0$), with normalized error in the upper panel.

Population spectral distributions.

Manipulation checks and individual differences.

Predictive modeling and brain–behavior.

Full fixed-effects table.

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

Model	Predictor	OR (95% CI)	p
Baseline (high grip)	Intercept	1.98 (1.60, 2.45)	$< .001$
Baseline (high grip)	Modality (visual vs. auditory)	1.70 (1.48, 1.96)	$< .001$
Baseline (high grip)	Stimulus difficulty (linear)	4.76 (3.86, 5.87)	$< .001$
Baseline (high grip)	Stimulus difficulty (quadratic)	30.19 (24.85, 36.68)	$< .001$

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

Model	Predictor	OR (95% CI)	p
Baseline (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	$< .001$
Baseline (high grip)	Stimulus difficulty (4th order)	1.75 (1.54, 2.00)	$< .001$
DSP-augmented (high grip)	Intercept	1.99 (1.61, 2.46)	$< .001$
DSP-augmented (high grip)	Modality (visual vs. auditory)	1.70 (1.47, 1.96)	$< .001$
DSP-augmented (high grip)	Stimulus difficulty (linear)	4.76 (3.86, 5.87)	$< .001$
DSP-augmented (high grip)	Stimulus difficulty (quadratic)	30.19 (24.85, 36.68)	$< .001$
DSP-augmented (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	$< .001$
DSP-augmented (high grip)	Stimulus difficulty (4th order)	1.75 (1.54, 2.00)	$< .001$
DSP-augmented (high grip)	Tracking power 0.5–3 Hz (z)	1.00 (0.93, 1.08)	0.965
DSP-augmented (high grip)	Tremor power 8–12 Hz (z)	0.96 (0.88, 1.04)	0.282

5 Reproducibility

Analysis notebooks live in `analysis/`:

Notebook	Purpose
`01_ingest.qmd`	Trial table construction and spectral feature extraction
`02_cognitive_motor_model.qmd`	Mixed-effects models and LC brain–behavior analyses
`03_hero_visuals.qmd`	Figure generation for manuscript and portfolio

Render from the repository root:

```
quarto render analysis/01_ingest.qmd
quarto render analysis/02_cognitive_motor_model.qmd
quarto render analysis/03_hero_visuals.qmd
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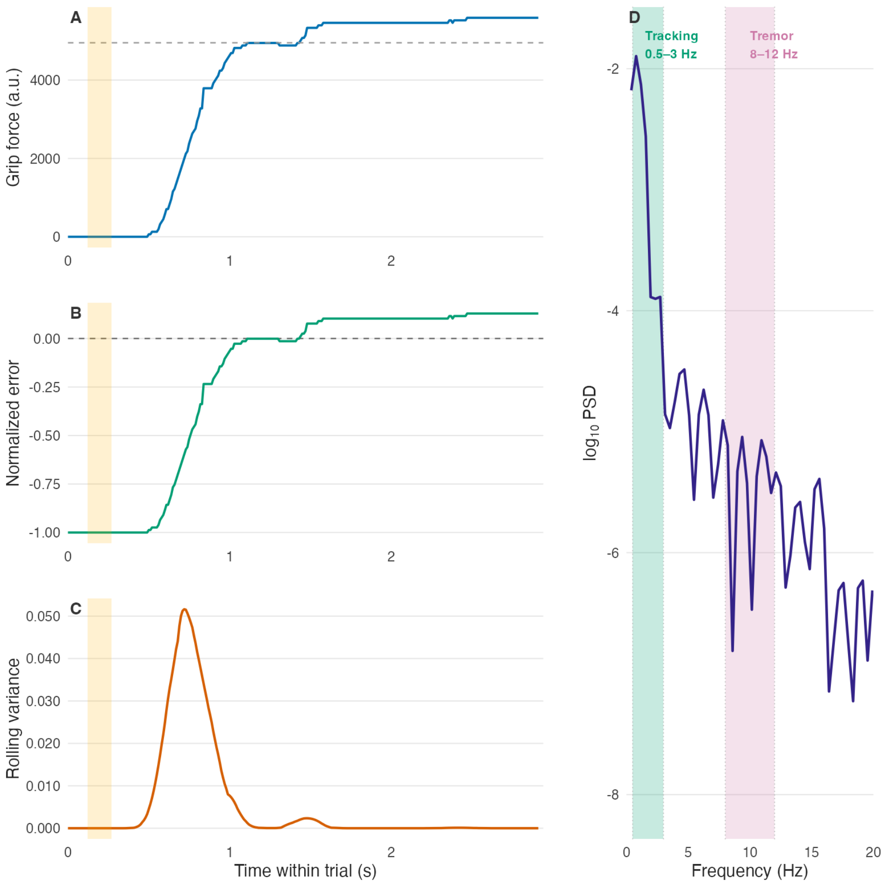

Figure 1: Single-trial signal chain from grip force to spectral phenotype.

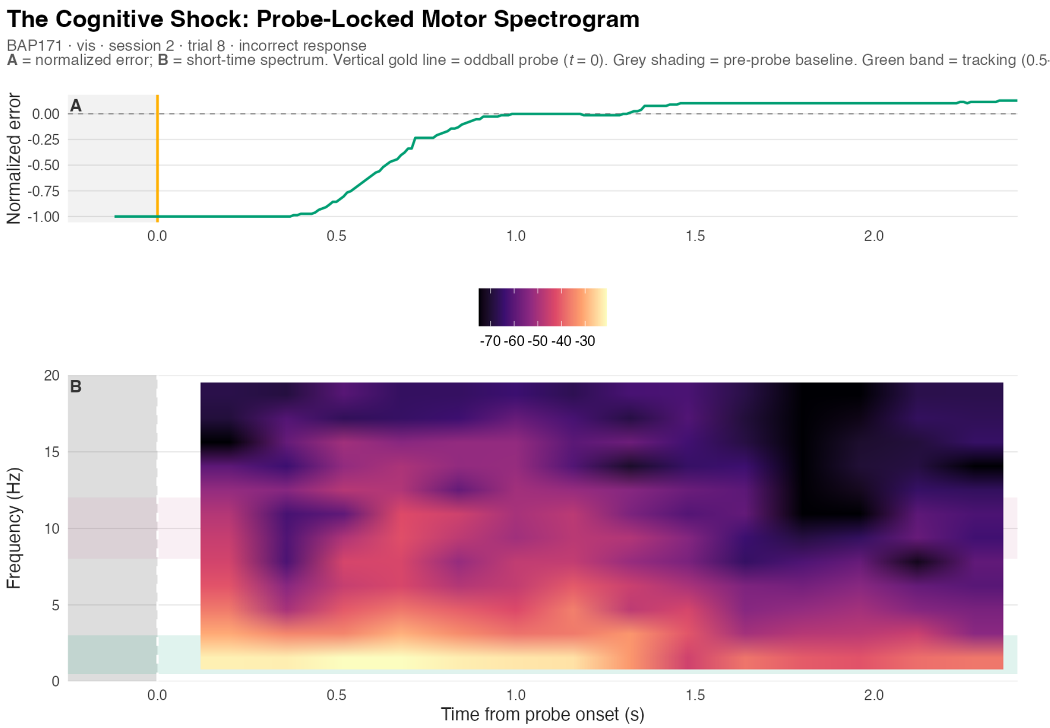

Figure 2: Probe-locked motor spectrogram aligned to oddball onset.

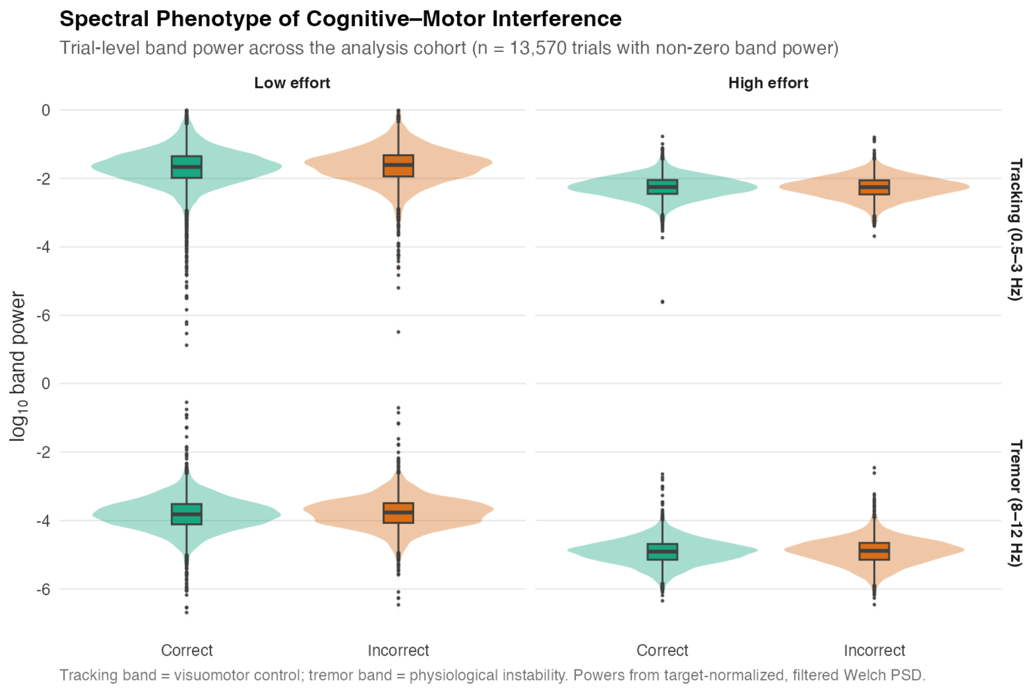

Figure 3: Trial-level log band-power by grip effort and perceptual outcome.

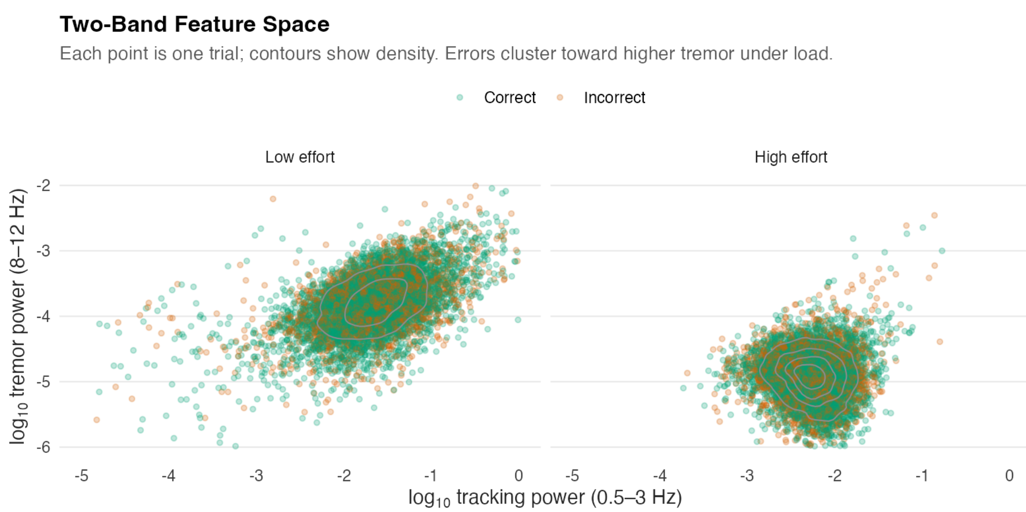

Figure 4: Two-band feature space (tracking vs. tremor), split by grip effort.

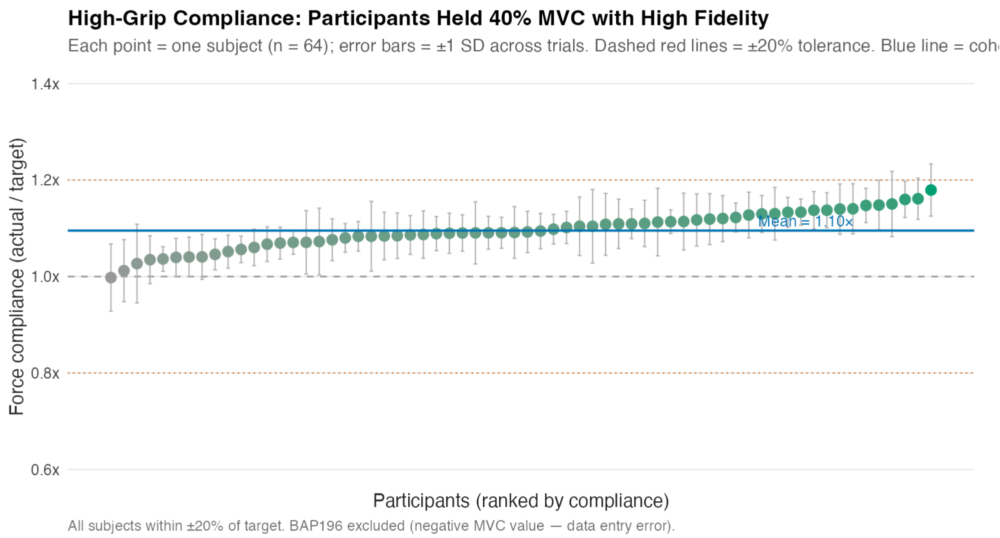

Figure 5: Per-subject high-grip force compliance (actual / target).

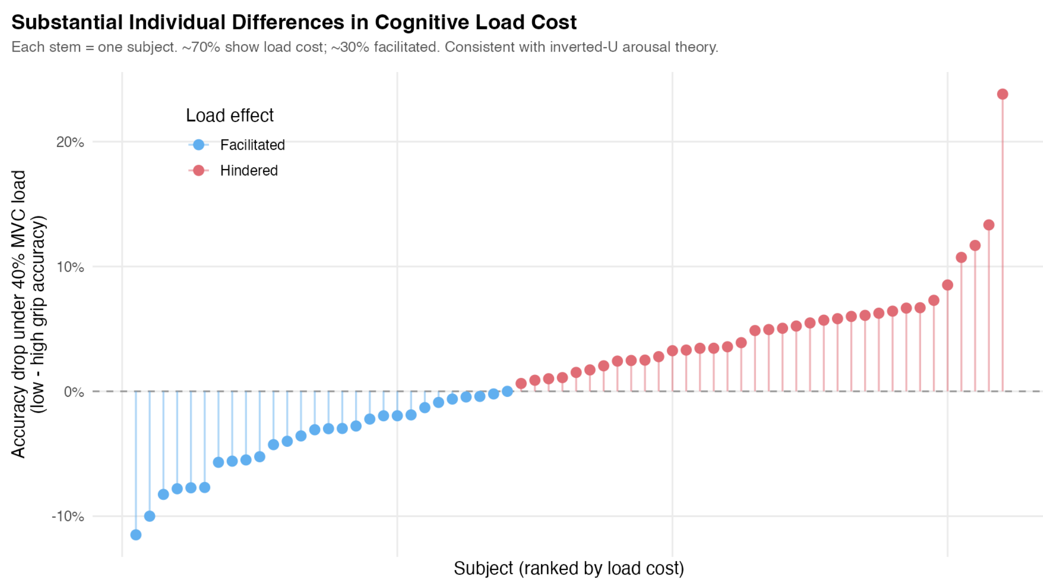

Figure 6: Subject-level accuracy drop under high versus low grip (caterpillar plot).

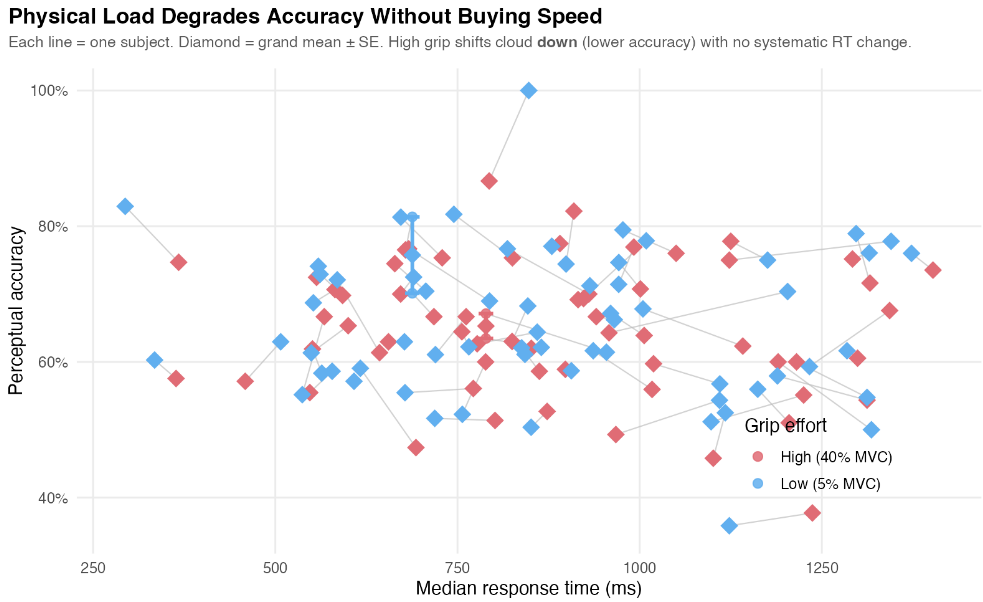

Figure 7: Speed–accuracy trade-off under low versus high grip.

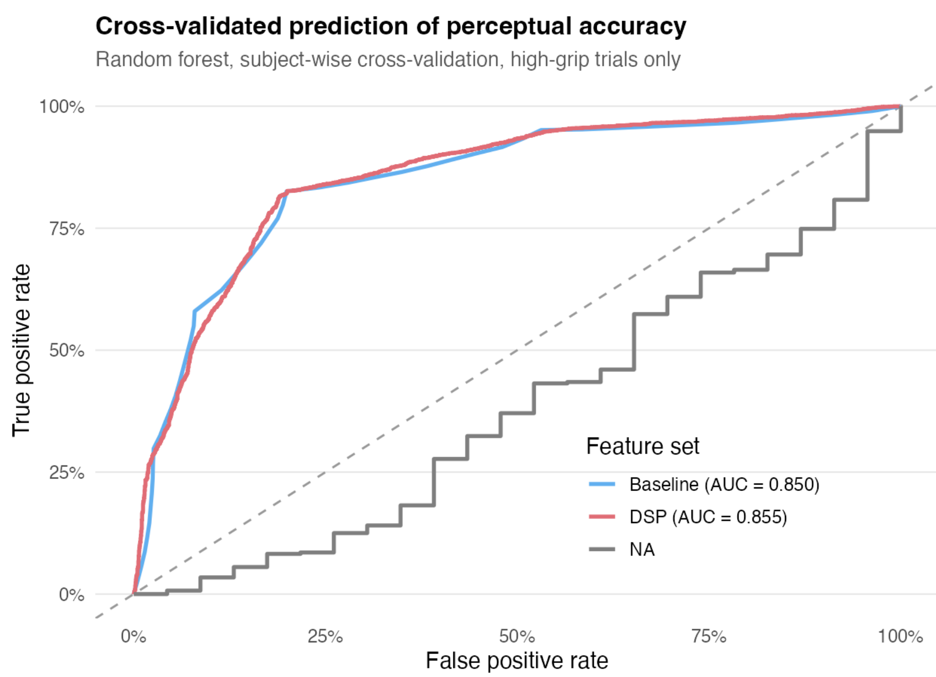

Figure 8: Cross-validated ROC curves (random forest, high-grip trials).

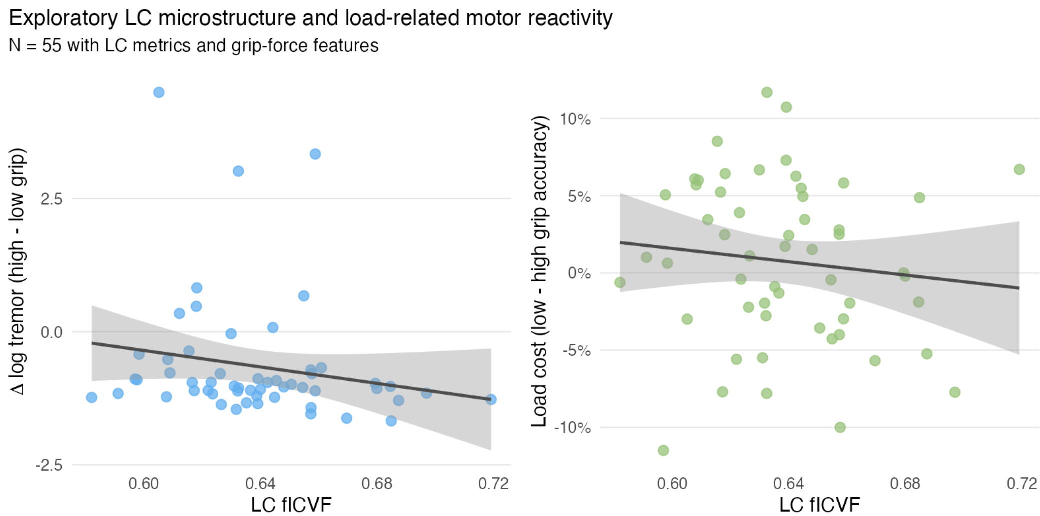

Figure 9: Exploratory LC fICVF associations with load-related tremor reactivity and perceptual load cost.

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quarto render manuscript/manuscript.qmd
quarto render manuscript/manuscript-biorxiv.qmd --to biorxiv-typst
quarto render manuscript/supplementary.qmd
```

Local data paths are configured in `config/paths.local.yaml` (not tracked in git).